Smart Agriculture AI — Comprehensive Farm Report

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1. Soil Analysis

Parameter	Value
Detected Soil Type	Clay
Detection Confidence	69.2%
Detection Method	EfficientNetV2 + SVM

2. Current Soil Conditions (IoT Sensor)

Parameter	Value	Status
Soil Temperature	31.4°C	Normal
Soil Humidity	54.0%	Normal
Soil Moisture	100%	Healthy
Irrigation Status	N/A	

3. Weather Conditions

Parameter	Value
City	Bhopal, India
Outside Temperature	33.6°C
Humidity	53%
Rainfall	0.0 mm/hr
Wind Speed	10.8 km/h

4. Field Details

Total Area: **10,000 m²** (2.47 acres)

5. AI Crop Recommendation

Mode: **Multi-Crop (Zone Allocation)**

Crop	Area (ac)	Alloc %	Yield (kg)	Price/kg	Profit (■)	Rating

Rice	1.23	49.6%	2,233	■15.79	■35,266	Excellent
Sugarcane	0.49	19.7%	17,702	■3.4	■60,187	Fair
Cotton	0.42	16.9%	423	■51.4	■21,732	Fair
Wheat	0.34	13.8%	517	■21.25	■10,986	Fair
TOTAL	**2.47**	**100%**	**20,875**		**■128,172**	

6. Crop Selection Reasoning

Rice (Score: 0.9):
• [OK] Compatible with Clay soil
• [OK] Temperature 31.4°C ideal (range: 20-35°C)
• [OK] Soil moisture 100% ideal (range: 60-100%)
• [i] Current rainfall: 0.0mm/hr

Sugarcane (Score: 0.36):
• [OK] Compatible with Clay soil
• [!] Temperature 31.4°C outside ideal range (20-30°C)
• [!] Soil moisture 100% outside ideal (60-90%)
• [i] Current rainfall: 0.0mm/hr

Cotton (Score: 0.31):
• [OK] Compatible with Clay soil
• [!] Temperature 31.4°C outside ideal range (21-30°C)
• [!] Soil moisture 100% outside ideal (30-60%)
• [i] Current rainfall: 0.0mm/hr
• [!] Higher risk crop (risk factor: 0.6)

Wheat (Score: 0.25):
• [OK] Compatible with Clay soil
• [!] Temperature 31.4°C outside ideal range (15-25°C)
• [!] Soil moisture 100% outside ideal (40-70%)
• [i] Current rainfall: 0.0mm/hr

Report generated by Smart Agriculture AI — IoT + Deep Learning System
Cost savings: Farmers typically spend ■10,000-20,000 on lab soil testing.